Experiment 1
Algorithm & Flowchart

Aim / Objective:

The aim of studying algorithms, flowcharts, and number systems is to build a strong foundation in logical thinking, problem-solving, and understanding how computers interpret and process data. Algorithms help in developing step-by-step logical solutions to problems, while flowcharts provide a visual representation of the program flow, making the logic easier to understand and communicate. Number systems are essential for understanding how data and instructions are represented and handled internally in computers. Together, these concepts form the core of programming and computational thinking, enabling students to design efficient, error-free, and well-structured programs.

Theory / Concept:

Algorithm:

An algorithm specifies a series of steps that perform a particular computation or task. An algorithm is 'an effective procedure for solving a problem in a finite number of steps. In mathematics and computer science, an algorithm is a step-by-step procedure for calculations. Algorithms are used for calculation, data processing, and automated reasoning. Starting from an initial state and given initial inputs it precedes in a stepwise manner to a final or desired state, producing "output" along the way. For example, a company may have a specific procedure to calculate the salary of its employees. The clerical staff who calculate the salaries repeatedly follow the process for each employee. They then hand over the salary slips and salaries.

Flowchart:

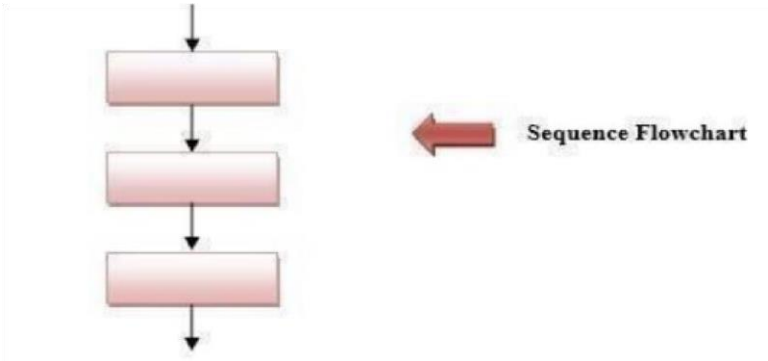
A flowchart provides the steps to be followed in order to arrive at the solution to a problem. It is a design tool used before the actual program is written. It may also be used as a communication tool to explain the logic to non-technical people. It is a diagram created using various standard shaped symbols. Given below is a list of basic shapes

Symbol	Purpose	Use
	Terminal(Stop/Start)	Start or end of a program or flowchart
	Input/Output	Used for input and output operation.
	Processing	Used for arithmetic operations and data-manipulations.

Symbol	Purpose	Use
	Decision	Used for decision making between two or more alternatives.
	Flow line	Indicates the flow of logic by connecting symbols.

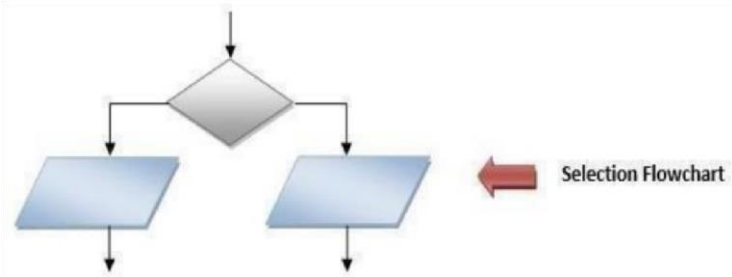
An Algorithm can have the following 3 types of constructs

1. Sequence: Sequence logic is used for performing instructions one after another in sequence.
- Sequence is the most basic of the constructs
- It is simply performing one step after another
- Each step is followed in a specific sequence, hence the name
- Sequence can be thought of as “do this, then do this, then do this”



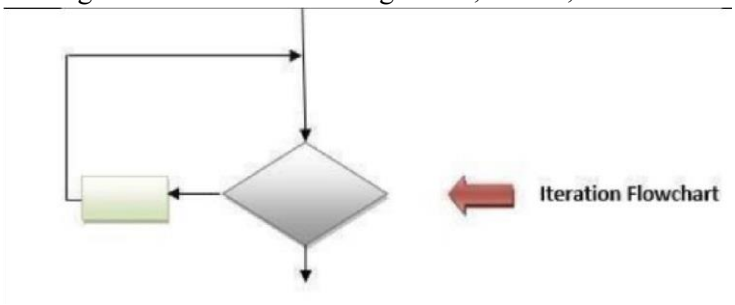
2. Selection: Selection logic, also known as decision logic, is used for making decisions.

- Selection logic is depicted as either an IF…THEN…ELSE or IF…. THEN structure. Selection is the decision-making construct.
- It is used to make yes/no or true/false decisions logically.
- Selection can be thought of as "if something is true, take this action, otherwise take that action".



Iteration: Iteration logic is also known as **Loop**. Iteration logic is used when one or more instructions may be executed several times depending on some condition.

- Iteration comes from the word "reiterate", which means to repeat
- Iteration is a looping construct
- Iteration is a combination of decision and sequence and can repeat steps
- Iteration can be thought of as "while something is true, do this, otherwise stop"



Problem Statement:

1. **Design an algorithm and flowchart to accept the distance between two cities in kilometers from the keyboard. Calculate and display this distance in meters, feet, cms, and inches.**

Algorithm:

Flowchart:

Code:

```
#include <stdio.h>
int main()
{
   float kms, meters, cms, feet, inches;
   printf("enter distance in kms : ");
   scanf("%f", &kms);
   meters = kms*1000;
   cms = meters*100;
   feet = meters*3.28024;
   inches = feet*12;
   printf("\nconverted into meters : %f m", meters);
   printf("\nconverted into centimeters : %f cm", cms);
   printf("\nconverted into feet : %f ft", feet);
   printf("\nconverted into inches : %f in", inches);
   return 0;
}
```

Output:

```
Output
enter distanace in kms :1

convrted into meters : 1000.000000 m

convrted into centimeters : 100000.000000 cm

convrted into feet : 3280.239990 ft

convrted into inches : 39362.878906 in


=== Code Execution Successful ===
```

2. **Design an algorithm and flowchart to accept a number from the user and find the remainder after dividing it by 2 and 3.**

Algorithm:

Flowchart:

Code:

```
#include<stdio.h>
int main()
{
   int n, rem1, rem2;
   printf("Enter a number : ");
   scanf("%d", &n);
   rem1 = n % 2;
   rem2 = n % 3;
   printf("Remainder when divided by 2 is %d\n", rem1);
   printf("Remainder when divided by 3 is %d\n", rem2);
   return 0;
}
```

Output:

```
C:\Users\Admin\Documents\(

Enter a number: 35
Remainder when divided by 2 =  1
Remainder when divided by 3 =  2

--------------------------------
Process exited after 4.131 seconds with return value 0
Press any key to continue . . .
```

Practice Questions:

1. **Design an algorithm and flowchart to calculate Simple Interest.**
2. **Design an algorithm and flowchart to find whether the input number is even or odd.**
3. **Write an algorithm for Movie Recommendation System.**

Learning Outcomes: Students will able to LO1
Define Algorithms and Flowcharts..
LO2 Illustrate the Algorithms and Flowcharts.

Course Outcome: After completion of course the students will be able to Algorithms and Flowcharts in C.

Conclusion:

In this experiment, we learned how to use Algorithms and Flowchart. They help in organizing thoughts, identifying logical errors, and simplifying complex problems before actual coding begins. By using algorithms and flowcharts, programmers can design more efficient and error-free programs, making these tools fundamental to the development and communication of effective software solutions.

For Faculty Use only

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	Total
Marks Obtained				